

● MEDIUM

● INFORMATION AND IDEAS

● ANSWER KEY

Command of Evidence

07

10 answers · Rationale-rich · Teach from doc

Q1**D**

- D. Each of the planets has a mass greater than 0.25 Jupiters, and all except for TOI-1478 b have an orbital period of less than 10 days.

Choice D is the best answer because it accurately describes data from the table that support Rodriguez and colleagues' assertion about the classifications of the five new gas exoplanets. The text describes two categories of gas planets: hot Jupiters, which have a mass of at least 0.25 Jupiters and an orbital period of less than 10 days, and warm Jupiters, which have the same mass characteristic but have orbital periods of more than 10 days. According to the table, four of the gas exoplanets discovered by Rodriguez and colleagues have a mass of at least 0.25 Jupiters and an orbital period of less than 10 days, while one of the planets has a mass of at least 0.25 Jupiters and an orbital period of more than 10 days. These data therefore support Rodriguez and colleagues' assertion that four of the new exoplanets are hot Jupiters and one is a warm Jupiter.

Choice A is incorrect because it doesn't accurately describe the data from the table. Although the table shows that TOI-628 b has a mass equivalent to 6.33 Jupiters, the table also shows that one of the planets—TOI-1478 b—does indeed have an orbital period of more than 10 days. Choice B is incorrect because it doesn't accurately describe the data from the table. Although the table does show that the masses of the five planets range from 0.85 to 6.33 Jupiters, the table also shows that TOI-1478 b has an orbital period of 10.180 days, not 153 days. Choice C is incorrect. According to the table, TOI-1333 b has an orbital period of only 4.720 days, not more than 10 days. Additionally, although the table does show that all the planets have a radius between 1.060 and 1.771 Jupiters, the text indicates that a planet may be classified as a hot Jupiter or a warm Jupiter based on its mass and orbital period, not on its radius, making the information about the range of the five planets' radius values irrelevant.

Q2

A

- A. “When I create playlists or choose songs to listen to, I often pick songs that I think the friends who are following me might be interested in listening to, rather than what I might actually prefer listening to in that moment.”
-

Choice A is the best answer because it directly reflects both aspects of Walsh’s finding about the social media dimensions of music streaming platforms: that they make their users more mindful of how their listening activities may be perceived and that self-awareness then influences how users manage those activities. The quotation illustrates this by showing that the user considers the music preferences of their followers when selecting music (“songs that I think the friends who are following me might be interested in”), even if those choices differ from their own preferences (“rather than what I might actually prefer listening to in that moment”). This supports both aspects of Walsh’s finding: users have increased social awareness and deliberately change their listening habits based on that awareness.

Choice B is incorrect because it focuses on how the platform’s algorithmic recommendation system influences user choice, not on how the imagined opinions of other users might influence that choice. The quotation highlights a user’s concern about repetitive music recommendations based on their past listening behavior. It does not suggest that the user is adjusting their listening habits based on how they think others may perceive what they listen to. Choice C is incorrect because it focuses on the social connection that can be created by music streaming rather than illustrating how social awareness affects users’ management of their listening activities. While this quotation addresses the social dimensions of streaming platforms, it describes how these features create shared experiences rather than demonstrating how concern about others’ perceptions influences users’ listening choices. Choice D is incorrect because it compares streaming platforms to traditional music purchasing methods. This comparison doesn’t address the social media dimensions of streaming platforms or how awareness of others’ perceptions of music selections influences what users choose to listen to.

Q3

B

- B. Compared with participants who saw the control review, participants who saw the low-anger review rated the review as slightly more helpful, whereas participants who saw the high-anger review rated the review as less helpful.
-

Choice B is the best answer. The claim is that low anger does not negatively affect readers' perceptions of the review, while high anger does negatively affect readers' perceptions of the review. This choice accurately expresses the supporting data from the "helpfulness of review" part of the graph: that low-anger reviews were rated as slightly more helpful than no-anger reviews, while high-anger reviews were rated as less helpful than no-anger reviews.

Choice A is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' ratings of the reviewed product are not relevant. Choice C is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' attitude towards the reviewed product is not relevant. Choice D is incorrect. This choice does not support the conclusion. The conclusion is only about how participants feel about the review itself—the participants' attitude towards the reviewed product is not relevant.

Q4

C

C. "I'm really getting quite fond of the big room, all but that horrid [wall]paper."

Choice C is the best answer because it most effectively illustrates the claim that the narrator of "The Yellow Wallpaper" has mixed feelings about her surroundings. She says she is "really getting quite fond of the big room," a positive sentiment, but also describes the room's wallpaper as "horrid," a negative sentiment. Since some of her feelings about her surroundings are positive and others are negative, they are best described as mixed.

Choice A is incorrect because though the narrator describes the room's wallpaper as "irritating," a negative sentiment, she does not mention a positive sentiment. Thus, the quotation does not effectively illustrate the claim that the narrator has mixed feelings about her surroundings.

Choice B is incorrect because it describes how the appearance of the room's wallpaper changes at night but does not mention the narrator's feelings about her surroundings. Choice D is incorrect because though the narrator describes the room's wallpaper as "repellant," a negative sentiment, she does not mention a positive sentiment. Thus, the quotation does not effectively illustrate the claim that the narrator has mixed feelings about her surroundings.

Q5

c

- C. Oulujärvi, which had fewer days of ice in the winter of 2005–06 than it did in the winter of 1980–81.
-

Choice C is the best answer because it most effectively uses data from the graph to exemplify the researcher’s claim—namely, that the duration of ice on some lakes has declined between the early 1980s and the mid-2000s. According to the graph, Oulujärvi had surface ice for nearly 200 days in the winter of 1980–81 but only about 160 days of ice in the more recent winter of 2005–06, evidence of a clear decline in the duration of surface ice between these time periods.

Choice A is incorrect because it focuses on only one period of time (the early 1980s). Thus, the data it cites don’t support the researcher’s claim, which compares the duration of surface ice across two specific periods of time (the early 1980s and the mid-2000s). Choice B is incorrect because although it accurately describes data from the graph, it contradicts the researcher’s claim about declining surface ice duration. It describes an increase, not a decline, in the duration of surface ice on Lake Neusiedl between the early 1980s and the mid-2000s (from about 80 days in 1980–81 to about 100 days in 2005–06). Therefore, the example of Lake Neusiedl wouldn’t logically support the claim that some lakes have seen a decline in the duration of surface ice. Choice D is incorrect because it provides information that doesn’t support the researcher’s claim about declining ice duration. The graph could indicate that both Lake Neusiedl and Oulujärvi had more than 105 days of ice in the winter of 2005–06, but on its own, this information doesn’t demonstrate a decline in ice duration between the early 1980s and the mid-2000s.

Q6

A

- A. Ancient guinea pigs in Puerto Rico were genetically less similar to ancient guinea pigs in Colombia than to ancient guinea pigs in Peru.
-

Choice A is the best answer because it presents a finding that, if true, would most directly weaken the underlined claim that guinea pigs in Puerto Rico probably came from Colombia and descended from Colombian guinea pigs. The text makes this claim based on the fact that ancient trade routes connected Puerto Rico with Colombia but not with Peru. However, if it's true that ancient guinea pigs in Puerto Rico were genetically less similar to ancient guinea pigs in Colombia than to ancient guinea pigs in Peru, this would suggest that the Puerto Rican guinea pigs didn't descend from Colombian guinea pigs. Instead, Puerto Rican guinea pigs' greater genetic similarity to Peruvian guinea pigs implies that ancient guinea pigs in Puerto Rico were likely brought to the island from Peru, despite the lack of direct ancient trade routes between the two regions.

Choice B is incorrect. While a finding about how common guinea pig imagery is in ancient Puerto Rican art would indicate the cultural significance of guinea pigs, it wouldn't be relevant to a claim about the geographical origin or descent of the guinea pig population on the island. The guinea pigs depicted in the artwork could have descended from guinea pigs in either Colombia or Peru; therefore, the finding would neither weaken nor strengthen the claim that guinea pigs in Puerto Rico probably came from Colombia and descended from Colombian guinea pigs. Choice C is incorrect because the finding that modern breeds of guinea pigs don't look like images of guinea pigs in ancient art from Puerto Rico, Colombia, and Peru wouldn't address the ancestral relationships between ancient guinea pig populations in these three regions. Therefore, the finding would neither weaken nor strengthen the claim that guinea pigs in Puerto Rico probably came from Colombia and descended from Colombian guinea pigs. Choice D is incorrect because a finding about the population sizes of guinea pigs in ancient Colombia and ancient Peru wouldn't be relevant to a claim about which population the Puerto Rican guinea pigs likely descended from. The smaller guinea pig population of Peru could still have been the source for the Puerto Rican guinea pig population.

Q7

D

- D. Grass cover increased from 1987 to 1993 and from 1993 to 2006, whereas forest cover decreased in those periods.
-

Choice D is the best answer because it describes data from the graph that support Martinez and colleagues' conclusion that the Jordanelle Dam led to changes that benefited grass plants but not trees. The graph shows characteristics of the banks of the Provo River downstream of the Jordanelle Dam in three different years—1987, 1993, and 2006. Specifically, the graph shows the amount of grass cover, bare soil, and forest cover in those years. The text indicates that the Jordanelle Dam was built in 1992, meaning that the data from the graph for 1987 reflect conditions before the dam was built, whereas the data for 1993 and 2006 reflect conditions after the dam was built. The data show that grass cover increased substantially from 1987 to 1993 and again from 1993 to 2006. The data also show that forest cover declined over those periods. Together, these data support Martinez and colleagues' conclusion that the dam was beneficial for grass plants but not for trees—grass cover increased significantly after the dam was built, while forest cover declined.

Choice A is incorrect. Although it is true that, in the graph, the lowest value for grass cover is approximately 58,000 square meters and the highest value for forest cover is approximately 75,000 square meters, both values are from 1987, before the Jordanelle Dam was built in 1992. Therefore, this information alone cannot support Martinez and colleagues' conclusion about changes in grass and tree cover following the construction of the dam. Choice B is incorrect because it presents an inaccurate description of data from the graph. The graph shows that there was more forest cover than grass cover in 1987, not that there was more grass cover than forest cover that year. Choice C is incorrect because, while it accurately reflects data from the graph when it compares grass cover and forest cover to bare soil, these data alone cannot support Martinez and colleagues' conclusion that the dam led to changes that benefited grass plants but not trees. An increase in grass cover relative to bare soil following the construction of the dam might indicate that the dam benefited grass plants, but the fact that there was more forest cover than bare soil in all three years doesn't indicate that the dam failed to benefit trees.

Q8

D

- D. Species that build open nests tend to have larger ranges than species that build domed nests.
-

Choice D is the best answer because it presents a finding that, if true, would challenge the assumption that many ecologists have made about the connection between the building of domed nests and geographic range in songbirds. The text says that many ecologists have assumed that since domed nests provide protection from weather conditions, songbird species that build such nests should be able to have larger geographic ranges than songbird species that build open nests do. If Medina and her colleagues found that species that build open nests tend to have larger geographic ranges than species that build domed nests do, their finding would show the opposite of what the ecologists have assumed. It would therefore challenge the ecologists' assumption.

Choice A is incorrect because nothing in the text suggests that there's a relationship between songbird species' extinction rates and their geographic ranges. The finding that species that build open nests tend to have higher extinction rates than species that build domed nests do would therefore have no clear bearing on the ecologists' assumption that domed nests allow species that build them to have larger geographic ranges than those of species that build open nests. Choice B is incorrect because nothing in the text suggests that there's a relationship between songbird species' sizes and their geographic ranges. The finding that species that build open nests tend to be smaller in size than species that build domed nests are would therefore have no clear bearing on the ecologists' assumption that domed nests allow species that build them to have larger geographic ranges than those of species that build open nests. Choice C is incorrect because although the text indicates that many ecologists have assumed that there's a connection between how songbird species build their nests and the species' geographic ranges, the text says that this assumption is based on the shape of the nests—that is, whether the nests are domed or open—not the number of materials used. The finding that species that build open nests tend to use fewer materials to build their nests than species that build domed nests do would therefore have no clear bearing on the ecologists' assumption that domed nests allow species that build them to have larger geographic ranges than those of species that build open nests.

Q9

C

- C. 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide, but only 16% believe that it is somewhat or very likely that a robot could work as a surgeon.
-

Choice C is the best answer because it effectively uses data from the table to complete the example of variations in participants' evaluations. The table shows participants' evaluations of the likelihood that robots could work effectively in different occupations. The text asserts that participants' evaluations varied widely depending on the occupation being considered and introduces an example supporting that assertion. The data from the table showing that 82% of participants believe that it is somewhat or very likely that a robot could work effectively as a tour guide but only 16% of participants believe that it is somewhat or very likely that a robot could work effectively as a surgeon illustrate this assertion: those data show participants' views changing substantially with the occupation being considered.

Choice A is incorrect because it does not describe data that illustrate the assertion that participants' evaluations varied widely depending on which occupation was being considered. Although this choice accurately describes data in the table, data about participants' responses when considering a single occupation—teaching—could not be an example of people's views changing substantially depending on the occupation being considered. Choice B is incorrect because it identifies a similarity in participants' responses when considering two different occupations, but the assertion that the example is intended to illustrate is that participants' evaluations varied widely depending on the occupation being considered. Although this choice accurately describes data in the table, those data do not illustrate the assertion in the text. Choice D is incorrect because it does not describe data that illustrate the assertion that participants' evaluations varied widely depending on which occupation was being considered. Although this choice accurately describes data in the table, data showing participants' evaluation of just one occupation could not be an example of participants' evaluations changing depending on the occupation under evaluation.

Q10

D

- D. Segments of the Missouri River flowing into lakes typically had dominant wavelengths of light above 560 nm, while segments flowing out of lakes typically had dominant wavelengths below 560 nm.

Choice D is the best answer because it presents a finding that, if true, would support Gardner and colleagues' conclusion that segments of the Missouri River flowing into lakes tend to carry more sediment than do segments of the river flowing out of lakes. The text says that rivers appear yellow when they contain a lot of sediment and appear red when they contain a lot of algae. It goes on to explain that Gardner and colleagues measured the wavelengths of light for different segments of rivers in the United States and classified those wavelength measurements into colors: red for wavelengths of 495 nanometers and below, blue for wavelengths between 495 and 560 nanometers, and yellow for wavelengths of 560 nanometers and above. Combined with the earlier information about river colors, this suggests that rivers rich in sediment will have wavelengths of 560 nanometers and above (since such rivers appear yellow). If researchers found that Missouri River segments flowing into lakes tend to have wavelengths above 560 nanometers and segments flowing out of lakes tend to have wavelengths below 560 nanometers, this finding would support Gardner and colleagues' conclusion, since it would suggest that the river tends to carry more sediment when it flows into lakes than when it flows out of lakes.

Choice A is incorrect because finding that sections of the Missouri River with high chlorophyll-a levels have wavelengths between 490 and 560 nanometers would be irrelevant to the researchers' conclusion that segments of the river flowing into lakes are richer in sediment than are segments of the river flowing out of lakes. This finding would not indicate anything about segments flowing into or out of lakes. Choice B is incorrect because finding that lakes through which the Missouri River passes have higher wavelengths near their shores than in the center would not support the researchers' conclusion that segments of the river flowing into lakes have more sediment than segments flowing out of lakes. This finding would suggest only that there is more sediment around the edges of lakes than in their centers, which does not have any direct bearing on the researchers' conclusion about river segments flowing into and out of lakes. Choice C is incorrect because finding that most segments of the Missouri River have wavelengths significantly higher than 560 nanometers would suggest that most segments of the river are high in sediment, not that segments flowing into lakes are higher in sediment than segments flowing out of lakes. Only a comparison of river segments flowing into lakes with segments flowing out of lakes can support the researchers' conclusion.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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